Mahmudul Haque

Software Engineer | C++, Computer Vision & AI

Chemnitz, Saxony, Germany mahmudulhoquetafhim@gmail.com

mhtafhim.github.io linkedin.com/in/mh-tafhim github.com/mhtafhim

Professional Profile

Incoming MSc Automotive Software Engineering student at TU Chemnitz with professional experience in C++, computer vision, machine learning, hardware-oriented desktop software, backend development, and requirements engineering. Conducted semiconductor R&D involving OpenCV image stitching, pattern recognition, spatial calibration, and 3D compensation. Strong in algorithms, object-oriented and modular design, and technical problem-solving, supported by 700+ solved programming problems and two first-place contest finishes. Highly comfortable with Linux/Unix systems.

Technical Skills

Programming: C++, C, C#, Python, Java, JavaScript, HTML, CSS
AI/Computer Vision: Computer vision, image stitching, geometric mapping, pattern recognition, LLMs, generative AI, deep learning, CNN, RNN, Deep Q-Networks
AI/ML Libraries: OpenCV, MediaPipe, DeepFace, RetinaFace, PyTorch, TensorFlow, CUDA, scikit-learn
LLM/Web/Backend: FastAPI, LangChain, ChromaDB, Ollama, LM Studio, React, ASP.NET Core, REST APIs, server-sent events
Cross-Platform/Systems: Qt, Linux/Unix, Fedora (daily use), Linux CLI, Windows, Docker, SSH
Desktop/Engineering: MFC, Visual Studio, GDI+, Java Swing, OOP, data structures, algorithms, modular design, code refactoring, software architecture
Build/Tools: Git, Visual Studio solutions/projects, .NET CLI, Docker Compose, Postman
Data/Cloud: PySpark, SQL Server, MySQL, Firebase, Apache Derby, AWS EC2, AWS S3
Simulation/Graphics: SUMO, OpenGL, iGraphics
Current interests: Cross-platform C++, embedded systems, CMake, multimodal AI, and real-time intelligent interactions.

Professional Experience

Software Engineer *Nov 2025 – Present*
Frontier Semiconductor (FSM), Dhaka, Bangladesh

- Conducted R&D on OpenCV-based image stitching to overcome camera field-of-view limitations, evaluating multiple geometric mapping, alignment, and blending algorithms to create seamless full-die images for inspection.
- Contributed to a pattern-recognition pipeline that located a selected reference feature across every wafer die, improving repeatable positioning and measurement accuracy; evaluated multiple recognition algorithms.
- Engineered a C++/MFC spatial mapping system to reconstruct physical semiconductor wafer die maps digitally and integrated the algorithms into an operator-facing production interface.
- Developed a four-point calibration algorithm across three adjacent dies to calculate and plot precise coordinates throughout the wafer radius.
- Modelled a virtual 3D plane from sampled focal points and predicted Z-axis compensation, resolving probe-focusing failures on tilted stages before measurement.

Business Analyst *Aug 2025 – Oct 2025*
US-Bangla Airlines, Dhaka, Bangladesh

- Led requirements sessions with cabin and cockpit crews, scheduling executives, and the Operations Control Center to map FlightOps processes.
- Translated stakeholder needs into structured technical specifications, software documentation, and UI/UX wireframes for developers.
- Designed backend business logic for crew training, including training cycles, certificate expiry, and programme assignments.

Education

MSc in Automotive Software Engineering *Starting Oct 2026*
Chemnitz University of Technology (TU Chemnitz), Chemnitz, Germany

BSc in Computer Science and Engineering *Jun 2021 – Jun 2025*
Ahsanullah University of Science and Technology, Dhaka, Bangladesh
CGPA: 3.719/4.00 **Credits:** 156 **EQF level:** 6 Degree record
Relevant coursework: Data Structures, Algorithms, Software Engineering, Data Analytics, Cloud Computing, Deep Learning, Pattern Recognition, and Machine Learning.

Projects

Face Match – Biometric Face Analysis *Python, OpenCV, DeepFace, MediaPipe, FastAPI* GitHub

- Built a computer-vision pipeline using RetinaFace alignment and FaceNet512 embeddings with cosine distance to determine whether two portraits show the same person.
- Extracted and visualised a 468-point facial mesh and normalised regional measurements for eyes, nose, mouth, jaw, face height, and eyebrows; exposed results through a FastAPI backend and React interface.

Local LLM Chatbot with RAG *Python, FastAPI, LangChain, ChromaDB, Docker* GitHub

- Developed a local LLM application with interchangeable Ollama and LM Studio providers, streamed responses, multimodal image/PDF attachments, and web-search augmentation.
- Implemented ingestion and retrieval for PDF, DOCX, text, and Markdown using LangChain chunking, provider embeddings, and persistent ChromaDB storage; packaged the services for Linux/Windows with Docker Compose.

Deep Reinforcement Learning for Intelligent Transportation Systems *Python, PyTorch, SUMO, DQN*

- Implemented and evaluated a DQN-based approach for dynamic route optimisation in urban traffic simulations.
- Analysed convergence and adaptability under changing traffic conditions, measuring congestion, fuel use, and emissions.

Reusable MFC Custom Controls Library *C++, MFC, GDI+, Visual Studio* GitHub

- Designed a modular library of 30 owner-drawn controls, including buttons, tabs, lists, sliders, status indicators, card panels, and a semiconductor wafer map.
- Separated reusable components from the demonstration application and documented APIs, integration patterns, resource dependencies, and known issues.

Data Analyzer *C++11, MFC, Visual Studio* GitHub

- Built a Windows engineering tool that parses structured text data, presents table and hierarchy views, and generates interactive charts with selectable axes.
- Organised file I/O, data vectors, tabbed analysis modules, chart controls, and image/data export into extensible application components.

Railway eService Platform & API *React, C#, ASP.NET Core 8, SQL Server* Web / API

- Developed a responsive ticketing platform and RESTful backend for route search, scheduling, bookings, fares, and user management with JWT authentication, password hashing, Swagger documentation, and SQL Server persistence.

SonicTheHedgeHog *C++, iGraphics/OpenGL* GitHub

- Built a 2D platformer with custom physics, frame-accurate collision handling, obstacle detection, and optimised rendering.

Achievements

- First place, Intra-AUST Programming Contest, Spring 2023 (100+ teams).
- First place, Intra-AUST Programming Contest, Fall 2022 (100+ teams).
- Khondaker Ahmed Muktadir and Mosammat Khodeza Begum Scholarship, Department of CSE, AUST.
- Fully funded scholarship for students from rural areas, awarded to the top 3% of students, Dhaka Ahsania Mission.
- Competitive programming: Codeforces 1263 (Pupil), CodeChef 1542 (3 Star), and 700+ problems solved across online judges.

Languages

Bengali: Native **English:** Fluent **German:** A1 (beginner)

Leadership & Volunteering

- **Head of Organising Panel, AUST IUPC 2025:** Directed a national inter-university programming contest.
- **Executive Member, AUST Programming & Informatics Club:** Organised programming contests, a mathematics olympiad, quizzes, and chess competitions.
- **Volunteer, Lions Clubs International (2018 – Present):** Supported flood relief, free healthcare camps, and schools serving street-connected children in Bangladesh.